MSC ARTIFICIAL INTELLIGENCE DISSERTATION

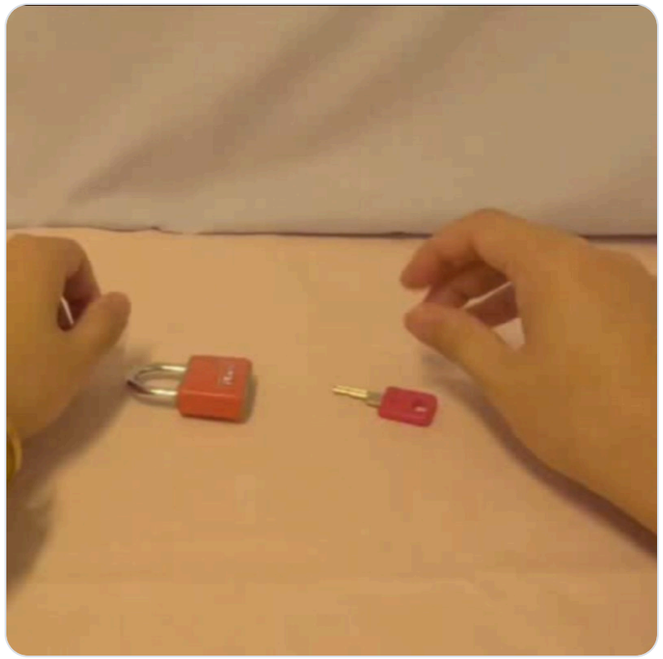
Learning to Select Actions for Complex Hand Tasks

A comparative study of visuomotor policies and latent world models

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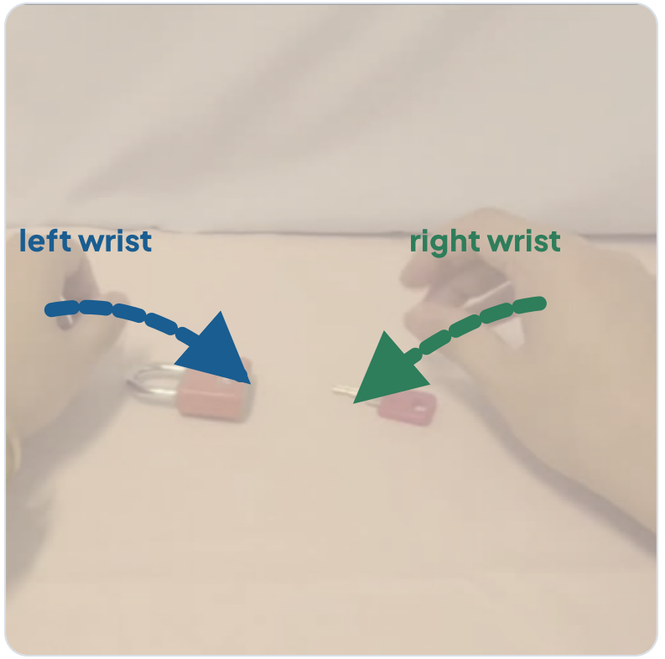
What is a policy?

A policy is a model that learns from demonstrations (human, robot or simulated) to turn the current scene into a chunk of future actions.



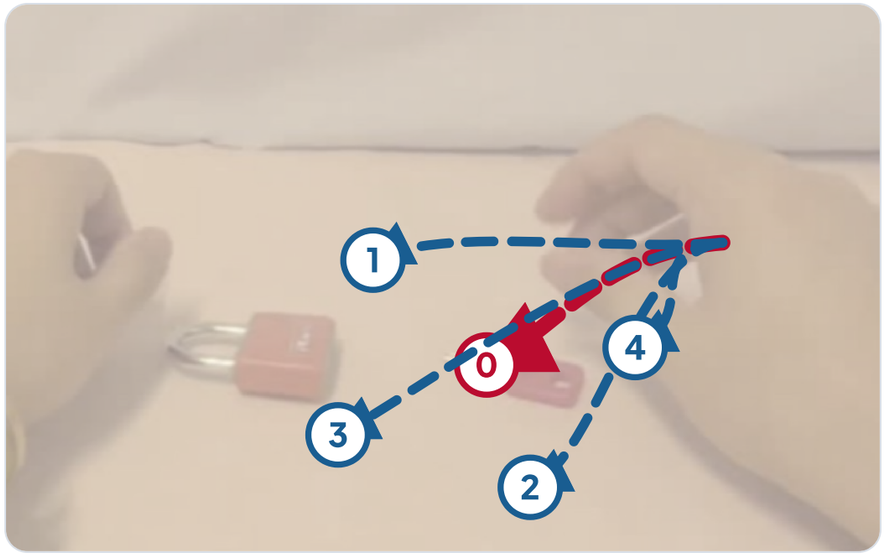
In this study, input: the current scene



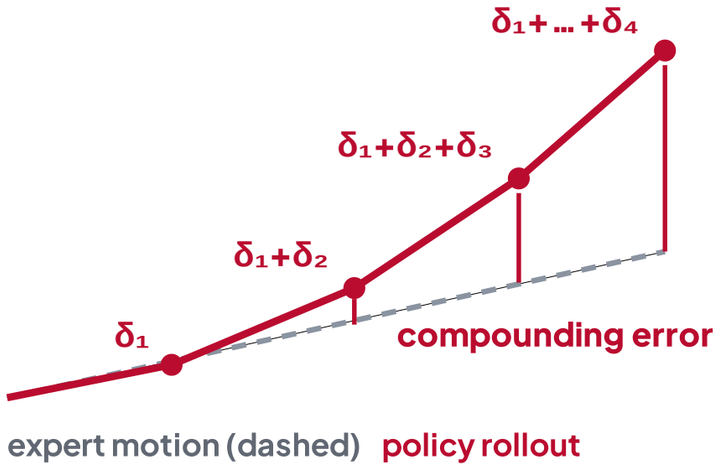


In this study, output: an action chunk, the next 16 wrist poses for both hands (18 numbers per step)

The problem of selection in a policy



Different latent or noise samples give different candidates.
Candidate 0 is the policy's default output: z = 0 for ACT, the first sample for Diffusion Policy.



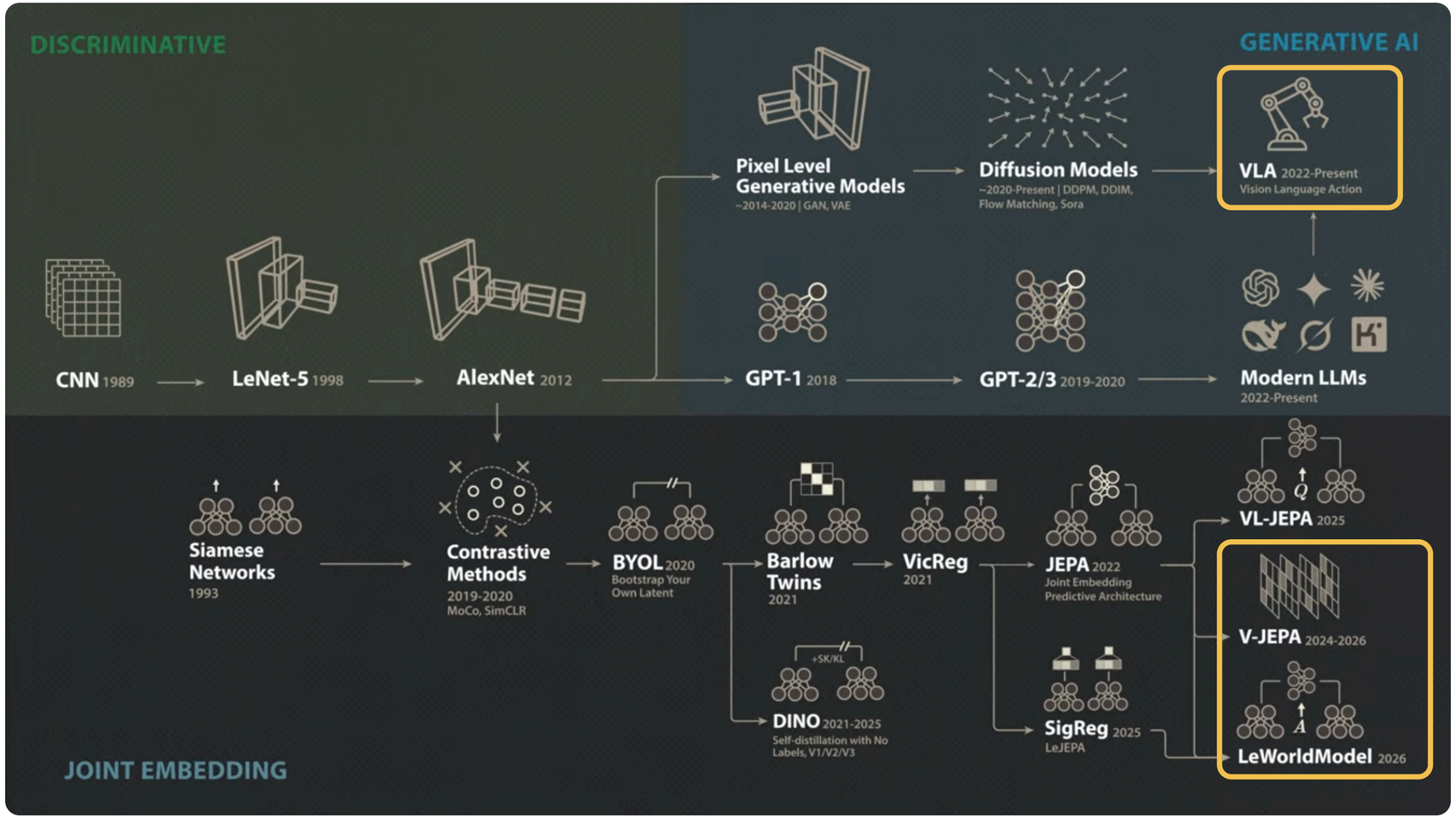
$$e_t = e_{t-1} + \delta_t = \delta_1 + \ldots + \delta_t$$

Suboptimal candidate → compounding error

→ unsuccessful or dangerous state

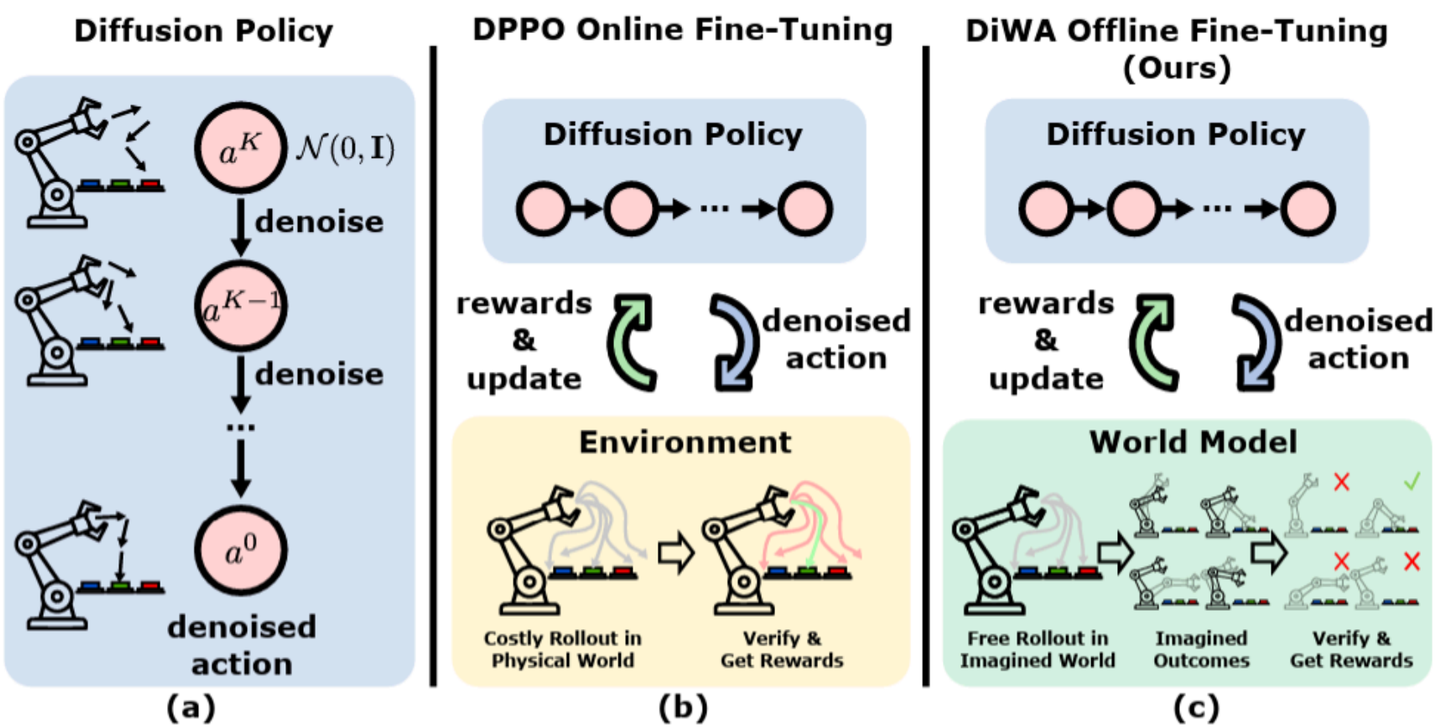
Why world models, not a VLA

A VLA maps the scene straight to an action, without predicting its consequences. A world model predicts those consequences first.

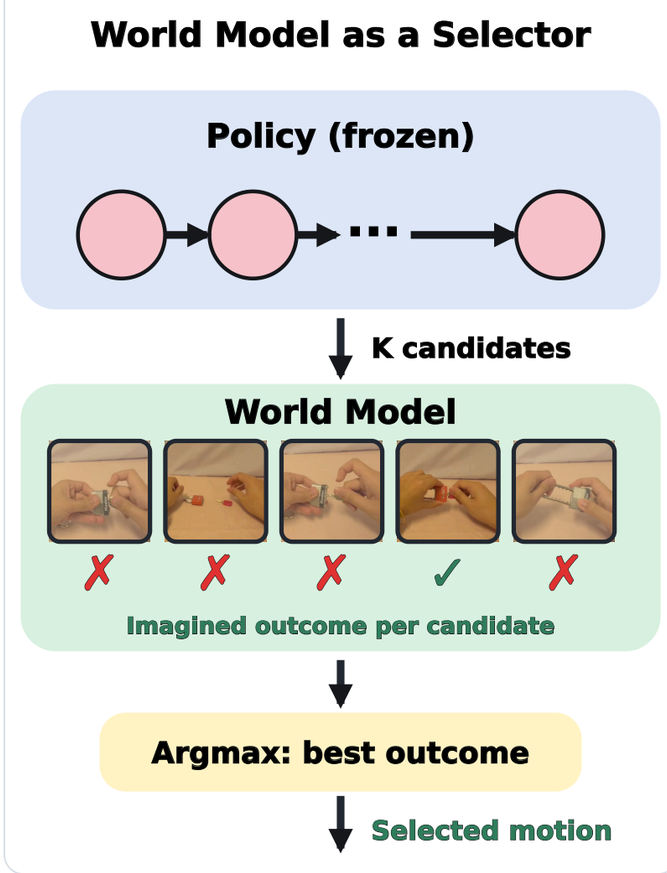


Screenshot from Welch Labs, Yann LeCun's $1B Bet Against LLMs (2026). Highlights added.

DIWA: THE PAPER WE BUILD ON



OUR APPROACH: EXTERNAL WORLD-MODEL SELECTION



Left: Chandra et al. (2025), evaluated on CALVIN. This study reproduces (c). Right: illustrative frames.

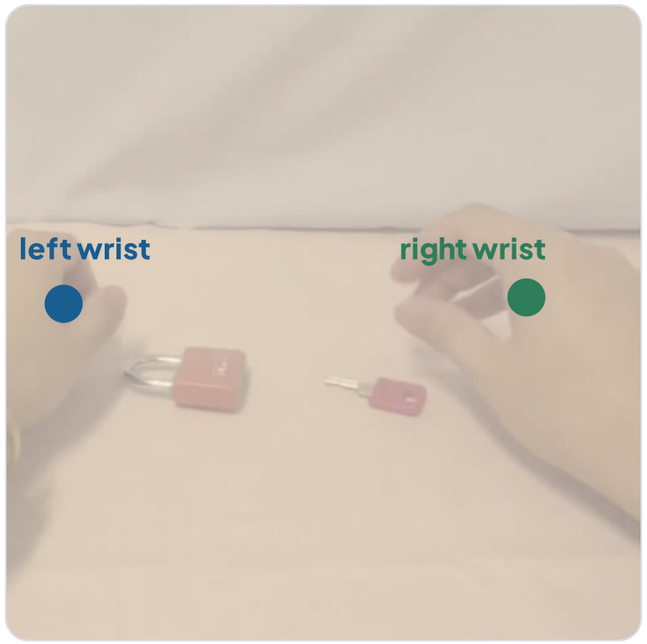
RESEARCH QUESTION

Can a frozen world model, used externally, pick a better motion from a policy's own candidates than the policy's single output?

Tested offline, with the world model acting only as the final decision stage over candidates the policy already proposed.

- How much does it help, and for which policy and world-model pairings?
- When does selection fail, and why?
- How does selecting compare with finetuning the policy inside a world model (DiWA on CALVIN)?

Dataset: EgoDex



Recorded with Apple Vision Pro, both hands tracked at 30 frames per second

EgoDex full release

829 hours, 338k episodes, 194 tasks

Training parts

1 2 3

Our training: parts 1 to 3

63 tasks, 194,333 episodes, 430,962 windows

4 5

not used

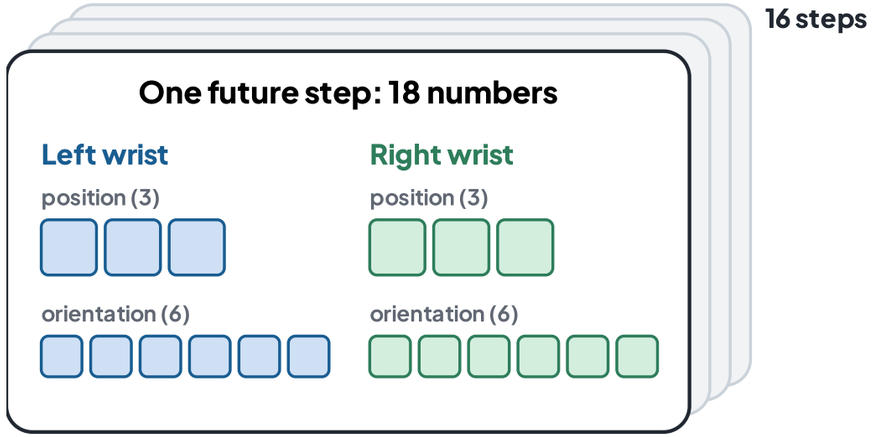
Official test set

111 tasks
3,229 episodes
7,607 windows

never used in training

Windowing scheme and data representation





18 numbers per step

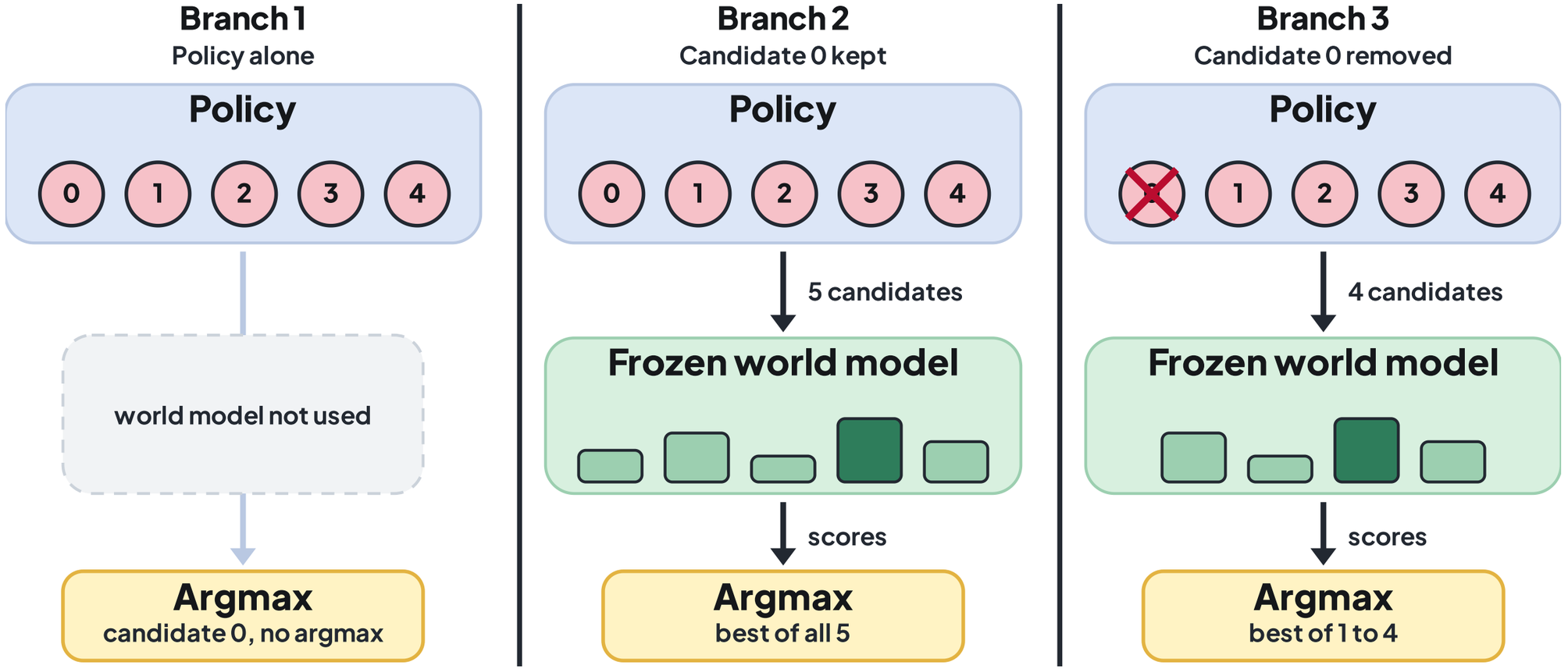
Per wrist: 3 position + 6 orientation (6-D rotation encoding)

12 DoF in total, 6 per wrist. Poses come from the headset's hand tracking.

430,962
TRAINING WINDOWS

7,607
TEST WINDOWS

THE METHOD



K = 5
CANDIDATES PER WINDOW

3 seeds
POOLED RESULTS

COMPUTE CONSTRAINTS

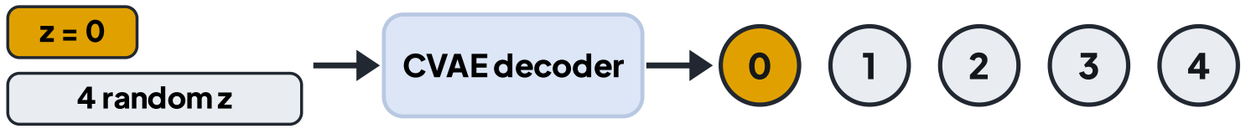
Understanding candidate selection

random, not tunable

Diffusion Policy
how it selects



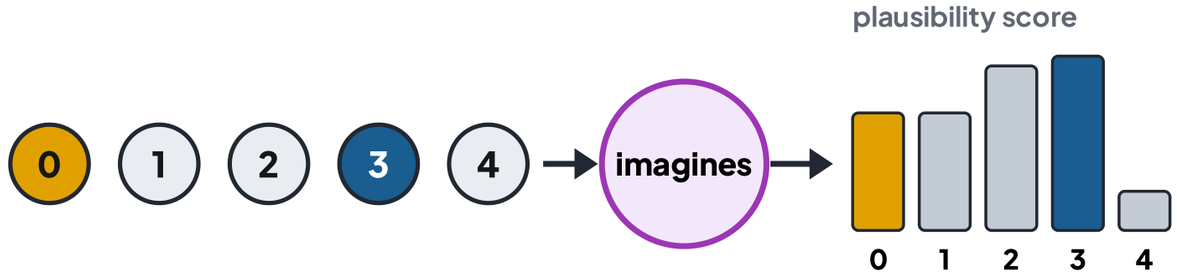
ACT
how it selects



No selection

Candidate 0 = the policy run once.
The first sample or z = 0,
and it is kept.

World model
how it improves it



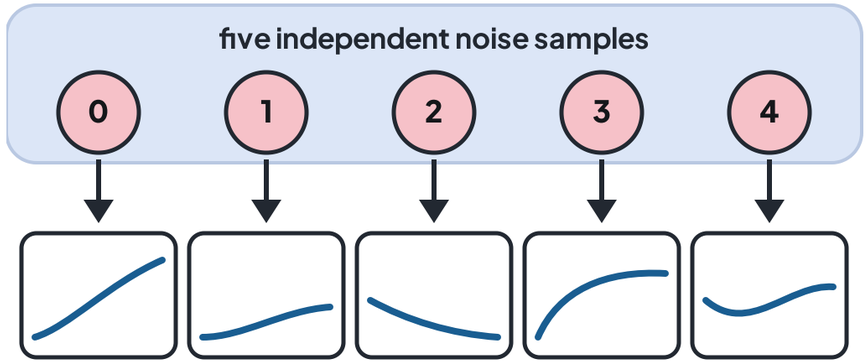
Selection

Candidate 3 has the highest plausibility score.

Never sees the real motion.
True error is measured afterwards.

Posterior collapse

Diffusion Policy

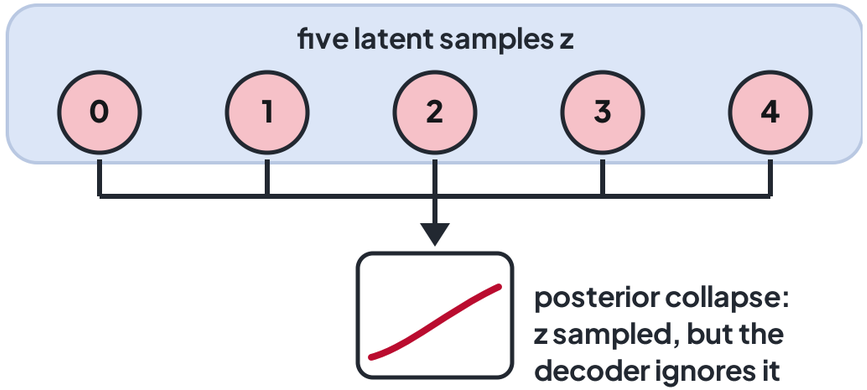


Five genuinely different motions

independent pool, clean measurement

Diffusion Policy denoises Gaussian noise into a motion over DDIM steps. 5 independent noise samples give 5 candidates.

ACT



Five copies of one motion

no diversity, so selection has nothing to choose between

ACT is a CVAE, where a latent z sampled at test time should vary the motion. Posterior collapse means the decoder learned to ignore z, so all 5 samples are the same motion.

ACT: four latent training configurations

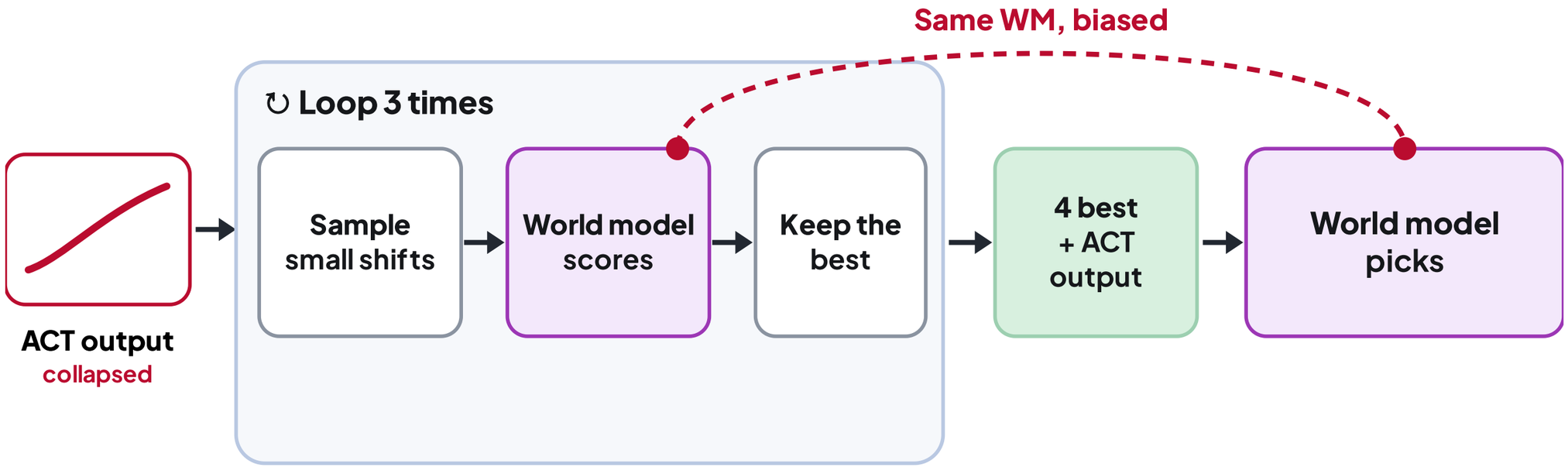
RUN	RECIPE	WHAT IT IS MEANT TO DO
v1	KL weight 10, the standard recipe	Baseline: the decoder can ignore z
v2	KL weight 100	Penalise information in z ten times harder
v3	Cyclical KL annealing and 30% observation dropout	Ease the penalty in cycles and hide observations, so the decoder must lean on z
v4	Cyclical KL annealing and free bits, 0.5 nats per dimension	Let z carry a minimum amount of information for free

No diversity

- All five samples identical
- Differences below 10^{-6}
- Posterior collapse

CEMP: Cross-Entropy Method with Policy

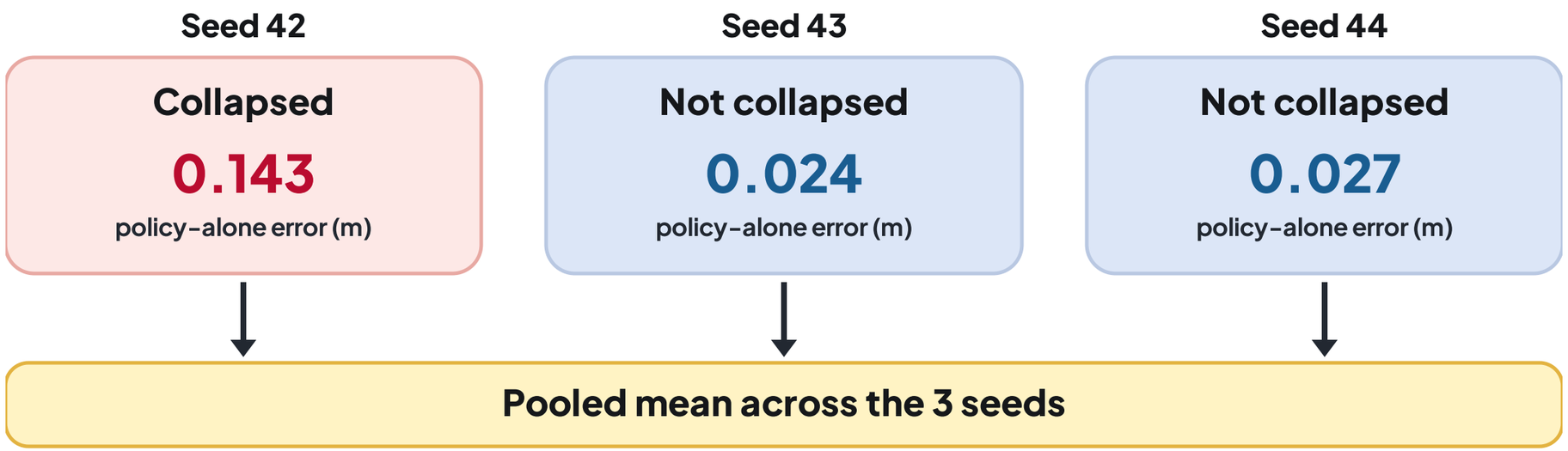
Cross-entropy method, started from the policy's output (inspired by TD-MPC2).



The same world model builds the pool and then picks from it, so the pool already leans towards what that world model prefers.

ACT results therefore measure the world model and the planner as one unit.

CEMP only applied where needed



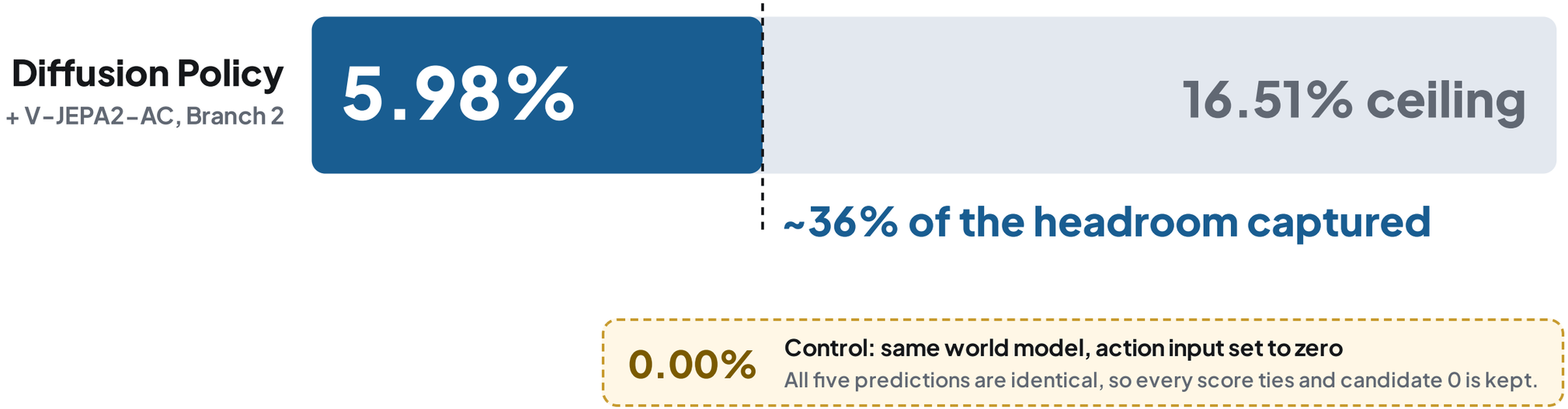
Only seed 42 needed CEMP, so seeds 43 and 44 were scored on ACT's own candidates. The pooled ACT result therefore mixes a world-model-plus-planner measurement with a plain scorer measurement. Read it as one result for this rule, not as a clean measurement of the scorer. This is why Diffusion Policy carries the main claim.

5 world models, 1 validated gain

MODEL	WHAT IT IS	GAIN	WHAT IT SHOWED
V-JEPA2-AC	Frozen video-pretrained encoder, our action head	-5.98%	Validated: its score tracks the true error
DexWM	Frozen DINOv2 encoder, second predictor and a hand-consistency head	+0.57%	No gain, single seed
LeWM	Small, compact, trained end to end, no pretrained encoder	-10.1%	Big gain, but its scores did not track the true error, so the gain is not from its predictions
V-JEPA2, no action	Control: same model, action input set to zero	0.00%	Every score ties: the motion is what matters
PointWorld proxy	Weakened motion input	+0.27%	No gain
DINO-WM, JEPA-WM	Official robot checkpoints, smoke-tested only	latent only	Interface mismatch: 18-D hand motion squeezed into 7-D robot actions

Translation error change, Diffusion Policy, Branch 2, matched training exposure. ACT with CEMP gave -0.58%, read as one unit (slides 12 to 14). LeWM detail: Backups C and D.

Ranking Diffusion Policy candidates with V-JEPA2-AC cuts translation error by 5.98%



The gain comes from reading each candidate's motion, not from the video backbone. Pooled across three seeds, 95% CI [5.43, 6.47], no retraining. Rotation error -0.96°. Win / tie / loss 49.9 / 23.2 / 27.0%. Branch 3: 5.37% of a 14.68% ceiling. The ceiling is what a perfect selector could remove from the same five candidates.

Secondary study on DiWA's own benchmark

C1
Policy alone

DiWA base policy

no world model

CALVIN simulator
success or fail

C2
Our external selection

DiWA base policy

5 candidates

World model

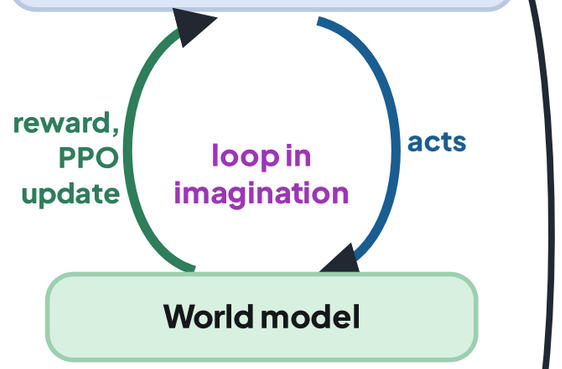


best one runs

CALVIN simulator
success or fail

C3b
DiWA finetuning, offline

DiWA base policy



World model

CALVIN simulator
success or fail

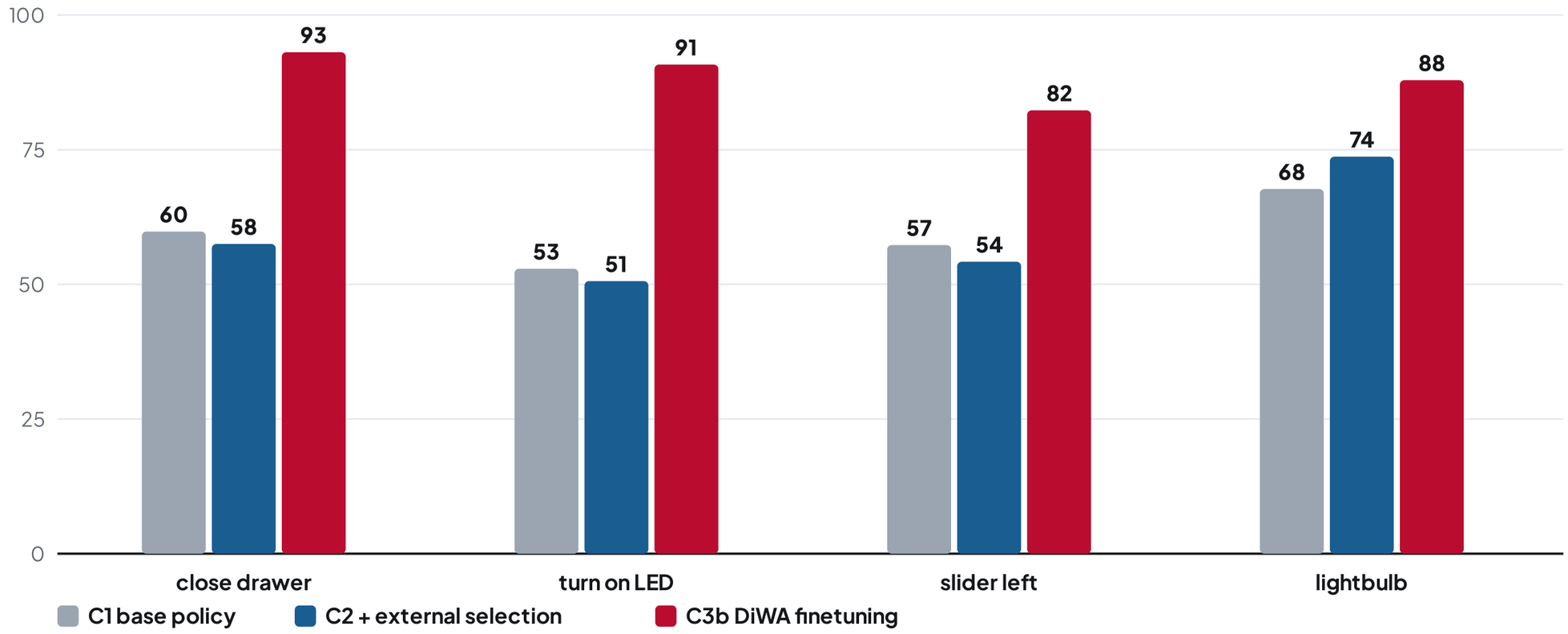
4 tasks
CALVIN MANIPULATION SKILLS

~100 episodes
PER TASK, ABOUT 33 PER SEED

3 seeds
42, 43 AND 44

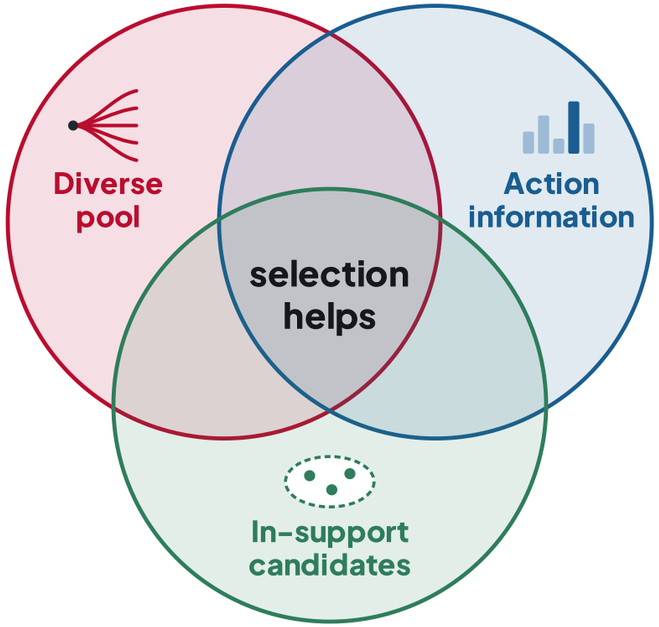
Success rate
% OF EPISODES COMPLETED

Finetuning > external selection



Task success rate (%), CALVIN. Selection (C2) is within noise of the policy alone (C1) on every task; finetuning (C3b) adds 14 to 40 points.

What a world model needs to help



Diverse pool

The candidates must be genuinely different motions, or there is nothing to choose between.

Action information

The score must depend on each candidate's motion. Without it, every score ties.

In-support candidates

Candidates must be close to the scorer's training data, so its predictions can be trusted.

Diffusion Policy + V-JEPA2-AC ● ● ●

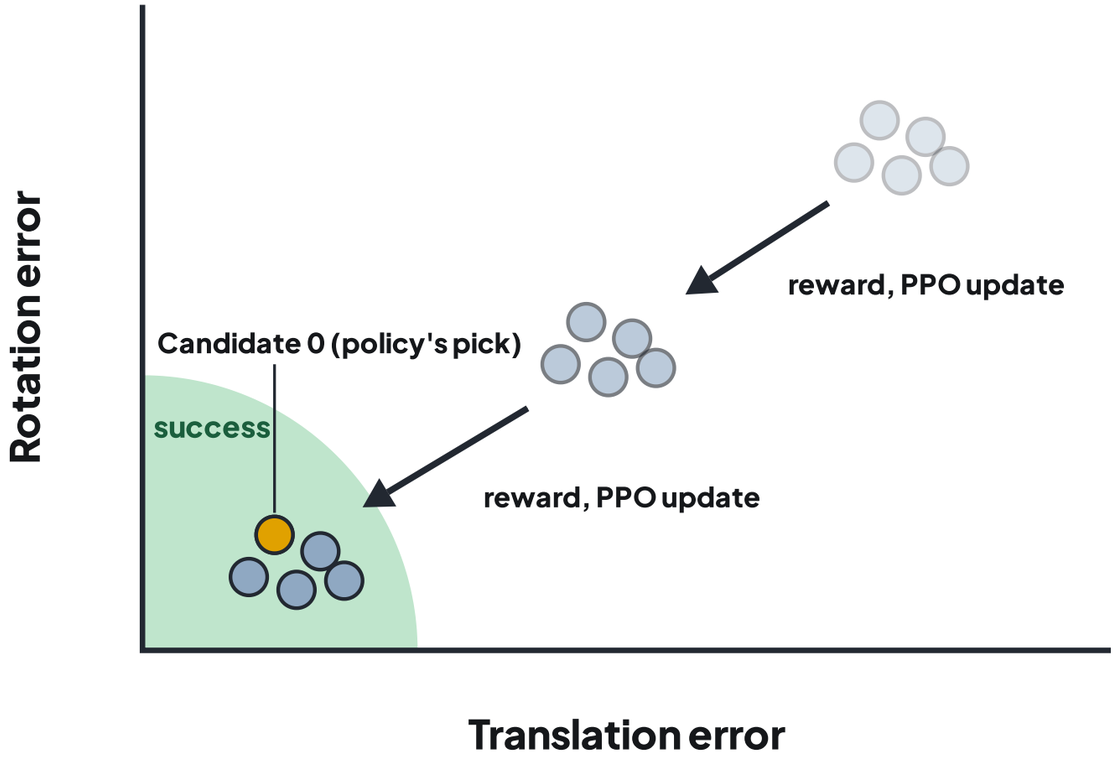
DiWA, by design ● ● ●

ACT, collapsed ○ ● ●

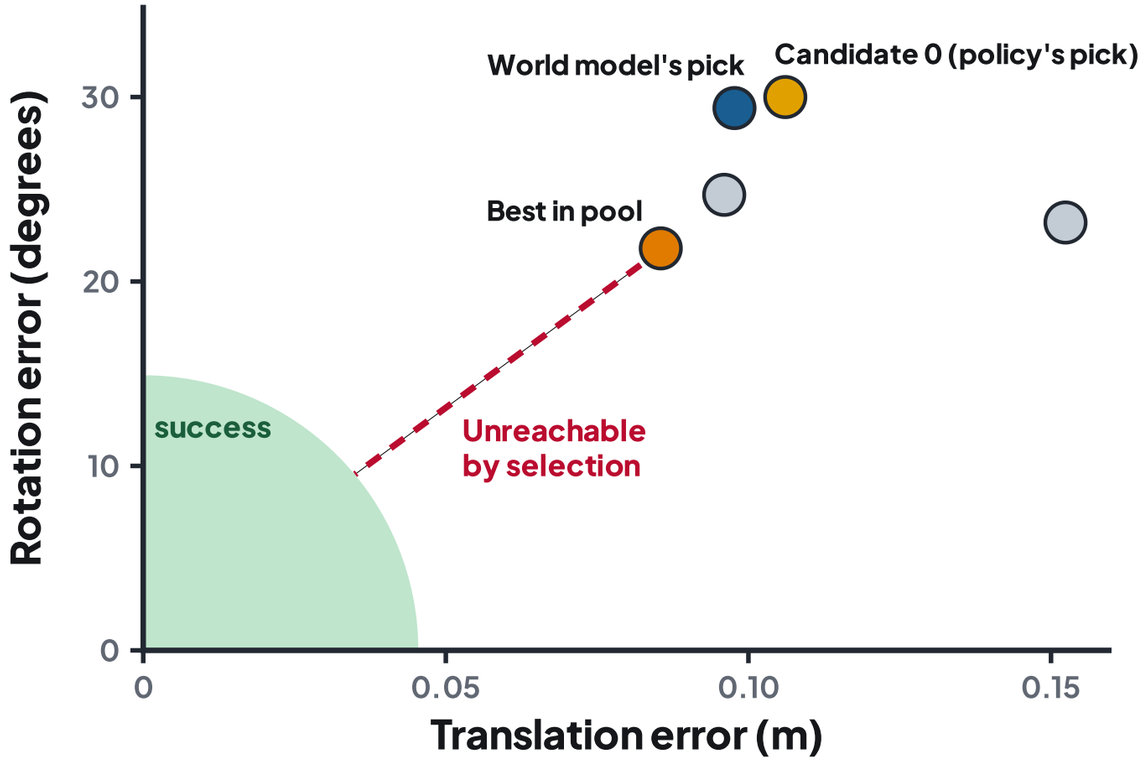
No-action control ● ○ ●

7-D adapters ● ● ○

DIWA (ILLUSTRATIVE)



OUR SELECTION



Finetuning improves the policy. Selection improves the choice.

Conclusions

1. **Yes, for one tested pairing**
   a frozen world model, used externally, selected better motions from the policy's own candidates

2. **Only when all three conditions hold**
   5.98% lower error, about 36% of the headroom, no retraining

3. **Finetuning moves the pool, selection picks from it**
   different tools, not rivals

Limitations

1

Time and compute

K = 5, three seeds, part of EgoDex

2

Offline and open loop

no robot or M.AI.ESTRO deployment yet, so compounding error is untested

3

Realistic, not successful

the scorer judges whether a motion looks realistic; we measured error against recorded human motion, never task success

Future work

1. **Close the loop**
   on a robot and with human guidance, in M.AI.ESTRO

2. **Combine finetuning and selection**
   with separate world-model judges, to get the best of both

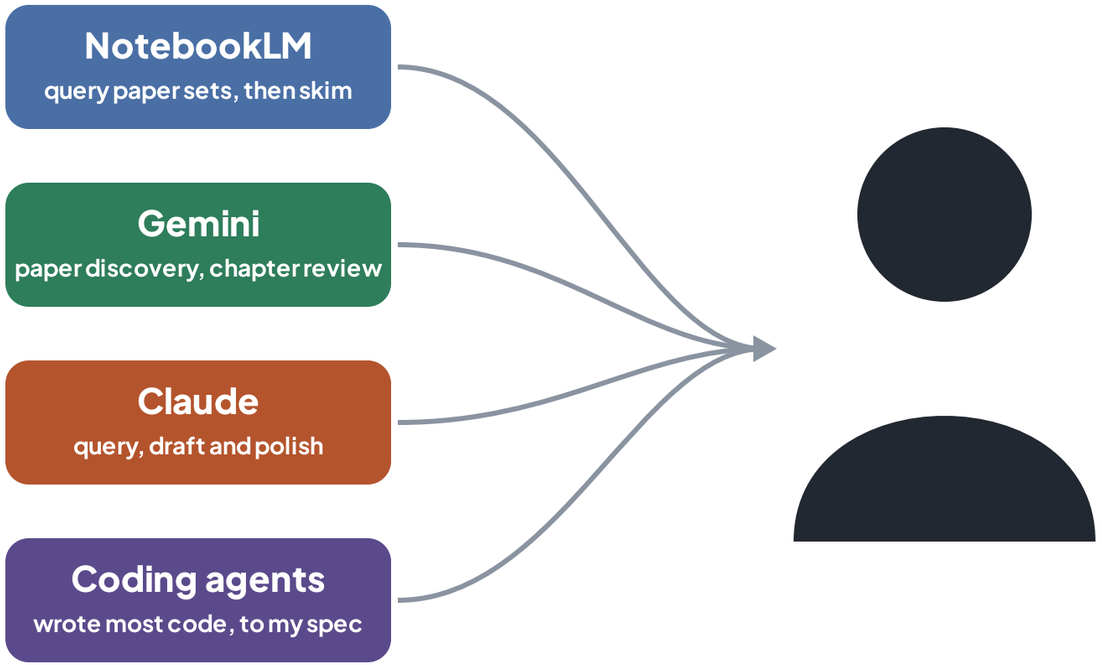
3. **Reach the ceiling**
   more data and stronger world models, to capture more of the 16.5% headroom

THANK YOU FOR LISTENING

Questions

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How AI was used in this work



My decisions

research question, benchmark design, which methods, what counts as a result, and every claim in the thesis

checked by audits and reruns

Audits found and fixed real defects before reporting, e.g. a DDIM sampler indexing error, a training-exposure undershoot and a DiWA configuration mistake.

Which published work anchored each choice

Diffusion sampling	→	DDIM (Song et al.), DP-T (Chi et al.)	Implemented
ACT collapse fixes	→	KL annealing (Fu et al.), free bits (Kingma et al.)	Tried, failed
Candidate search for ACT (CEMP)	→	Cross-entropy method, inspired by TD-MPC2	Adapted
Action-conditioned scorer	→	V-JEPA2-AC (Assran et al.)	Head trained locally
Finetuning comparison	→	DiWA (Chandra et al.) on CALVIN	Reproduced
K = 5, 100,000-item exposure	→	No paper: compute budget	Compute choice

LeWM: one model, two ways to score

Trained once, end to end from scratch on EgoDex (encoder, predictor, action embedder, projector)

Rollout scorer

predicts the future through its dynamics, then distance to real training latents

-10.09%

≈

Action-prior scorer

no prediction at all: how typical the motion is against training actions

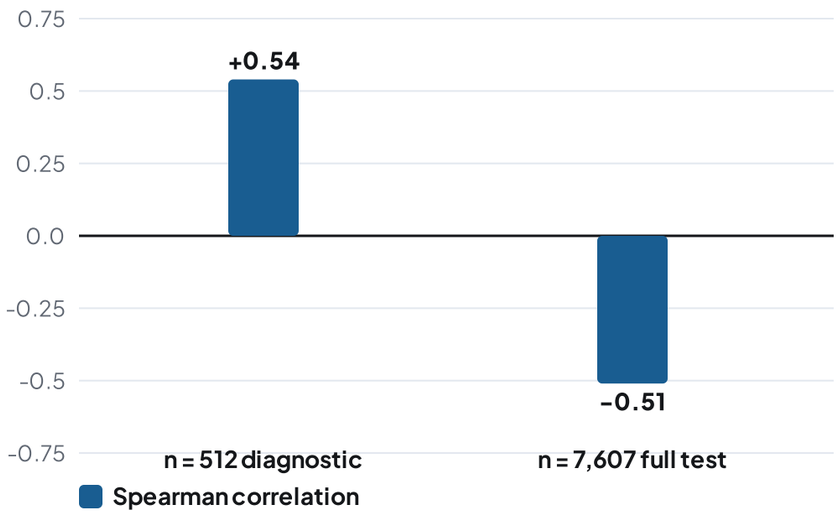
-10.24%

Fixed along the way:

frame-cap mismatch (97.9% of test windows dropped) · test-set action statistics (1.74× scale) · reweighting to pass the gate rejected

The two scorers give almost the same gain, and only the rollout scorer uses LeWM's learned dynamics. So the gain comes from motion regularity, not prediction.

LeWM's score does not track the true error



+0.54 on 512 windows, −0.51 on all 7,607

The −10.1% gain is real, but it does not come from LeWM predicting which candidate is better.

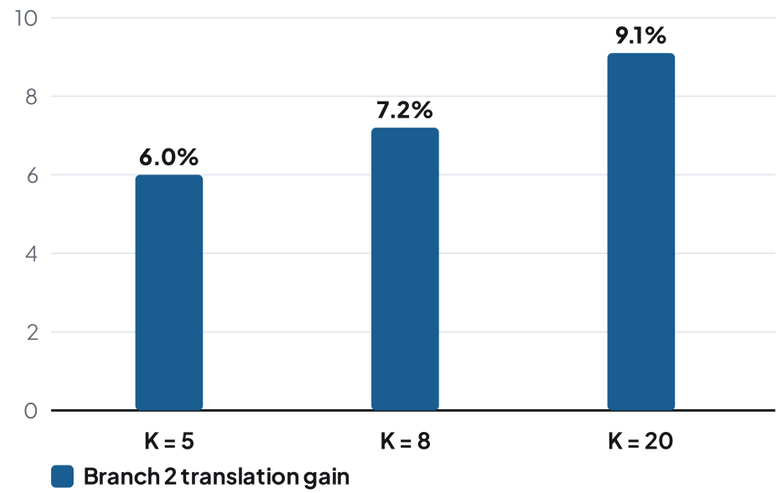
Spearman correlation between score margin and true translation improvement. Held out of the main comparison for this reason.

The result holds at every seed, and the controls behave as designed

	SEED 42	SEED 43	SEED 44
Baseline translation error	0.1142	0.1178	0.1191
V-JEPA2-AC, Branch 2	-6.41%	-6.09%	-5.46%
V-JEPA2 without action input	0.00%	0.00%	0.00%
PointWorld proxy	+0.50%	+0.23%	+0.07%
LeWM rollout (unvalidated)	-10.10%	-10.06%	-10.12%

Three seeds are a basic robustness check, not proof of stability. DexWM is single-seed by a cost decision of roughly 121 extra GPU-hours.

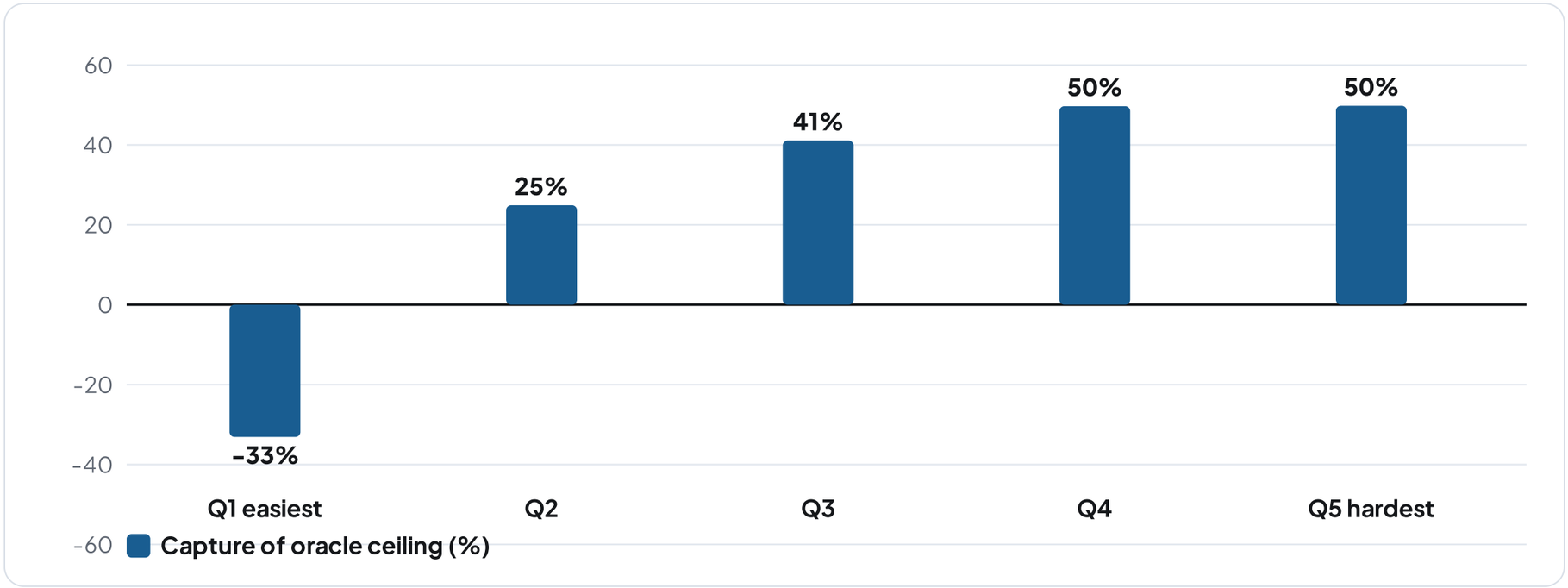
More candidates buy more headroom, not better selection



WHAT THE SWEEP SHOWS

- Scorer ranking identical at every K from 2 to 20
- Gain grows steadily with K and does not plateau
- Capture of the oracle ceiling stays flat, near 40% on these nested pools
- K = 5 was a compute choice, with no literature anchor

On the easiest fifth of windows, selection makes things worse



Windows sorted by baseline error, K = 8, Branch 2. Translation error rises 5.81% in Q1. Confidence gating does not help, because the score gap has a correlation of -0.002 with baseline error. A state-change-targeted windowing removes most of the reversal.

How results were measured

ERROR METRICS

- Translation, root-mean-square per axis, not Euclidean distance
- Rotation, mean geodesic angle in degrees
- Both averaged over the full 16-step chunk
- Change reported relative to candidate 0

STATISTICS

- 95% episode-clustered bootstrap, 2,000 resamples, pooled over seeds
- Oracle ceiling, best candidate in the same pool
- Ranking-validity gate, per-window Spearman correlation
- Win, tie, loss against candidate 0

Test windows are fixed, candidate pools frozen, future-goal access blocked, and scorer support restricted to training data.

What is disclosed about test-set use

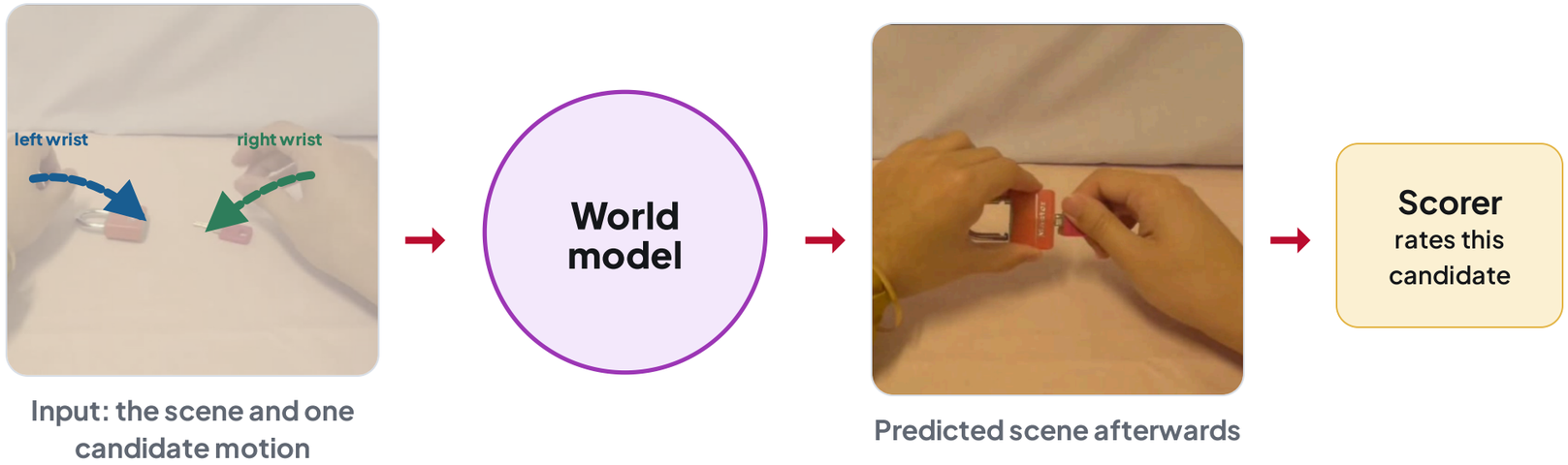
- The Branch 2 margin threshold's final value came from a disjoint validation split.
- An earlier, superseded calibration touched the test set directly. This is stated as a named limitation and is reassuring, not a full defence.
- The LeWM gate was tuned on a small diagnostic split and failed to generalise to the full test set. It was reported, not re-tuned.
- Earlier stronger-looking results relying on shortcuts, transductive support or oracle access were corrected rather than reported.

Objective achievement

OBJ.	STATUS	BASIS
1	Fully	Three-branch benchmark implemented as designed and used for every family reported
2	Partly	Local exposure matched, externally pretrained backbones carry an advantage not equalised
3	Within tested configurations	Whether and when selection helps determined for every combination, one positive, a likely explanation for each other

Objectives 4 and 5 are in the Conclusion chapter's objective table. Every non-positive-route explanation is the study's own interpretation, not independently proven.

A world model predicts, a scorer chooses

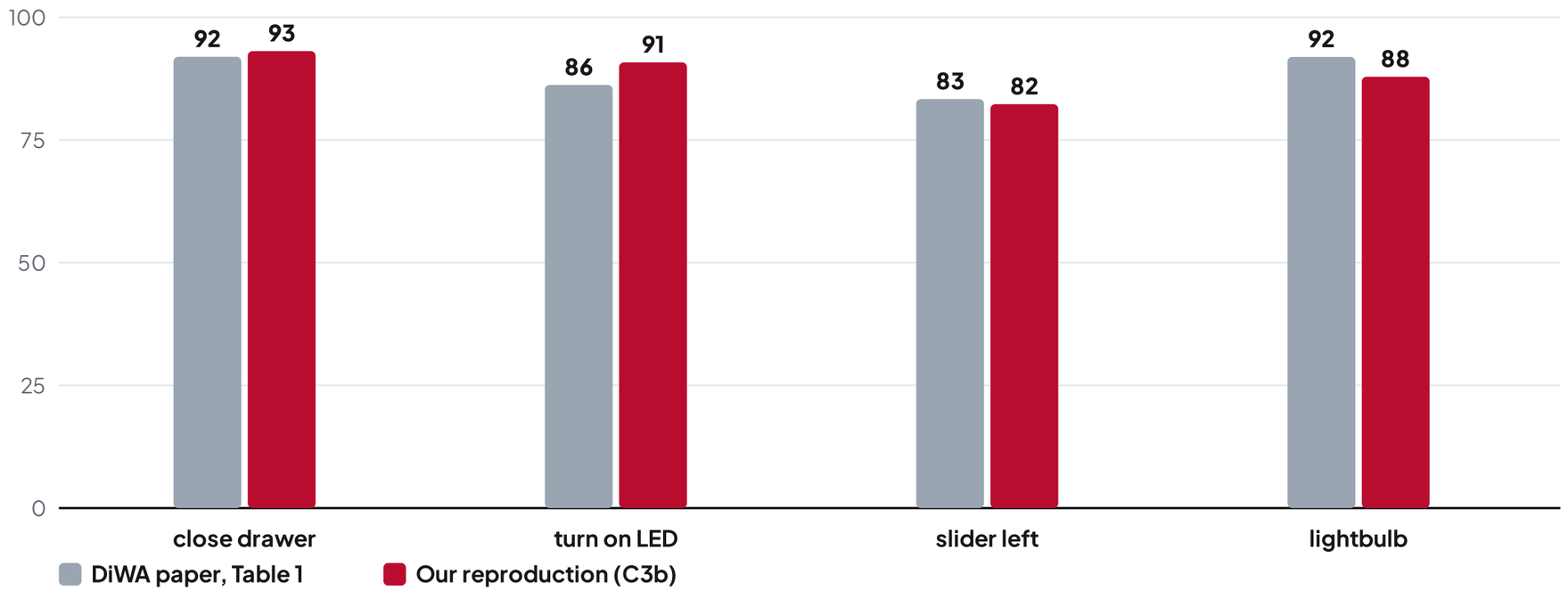


Input: the scene and one candidate motion

Predicted scene afterwards

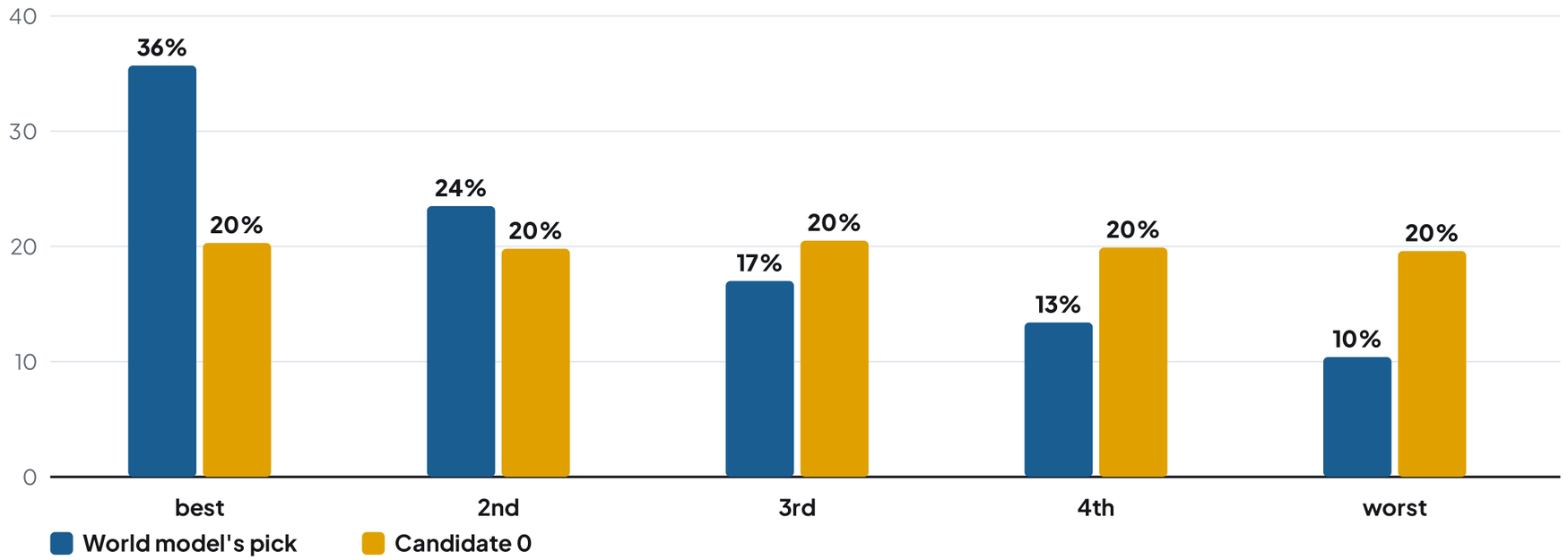
Score = -(0.25 x distance to the nearest real training future) - (0.1 x how unusual the motion is) - (0.05 x distance from candidate 0). The world model does not decide what to do. A prediction is not a decision, so a scorer must turn each predicted future into a score. Illustrative frames: the real prediction is made in a compact latent space, not in pixels.

We reproduced DiWA's finetuning



Success rate (%), CALVIN, three seeds, the authors' released checkpoints. Within 5 points on every task; base policy within 7 (Q&A 3). The behaviour-cloning weight was set to the paper's values after a run at zero collapsed two tasks.

Where each pick ranks among the five



Seed 42, 7,607 windows, ranked by true translation error. Candidate 0 is spread evenly, like a random pick. The world model finds the best candidate about 1.8 times as often as chance.

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